

ICS

English version

Railway applications - Track - Acceptance of works - Part 3: Acceptance of Rail Grinding, Milling and Planing Work in Track

Applications ferroviaires - Voie - Réception des travaux -
Partie 3: Critères d'acceptation des travaux de meulage,
frainage et rabottage des rails en voie

Bahnanwendungen - Oberbau - Abnahme von Arbeiten -
Teil 3: Annahme von Schleif-, Fräs- und Hobelarbeiten an
Schienen im Gleis

This draft European Standard is submitted to CEN members for enquiry. It has been drawn up by the Technical Committee CEN/TC 256.

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Foreword

This European Standard was established by the Technical Committee CEN/TC 256/SC 1 „Railway Applications/Track“, the secretariat of which is held by IBN.. It will be submitted to CEN members for formal vote.

((Include „Mandate“ - text))

Experts from the following countries were engaged in the drafting work:

Austria, Belgium, Czech Republic, France, Germany, Portugal, Spain, Switzerland, the United Kingdom.

This series of ENS „Railway applications - Track - Acceptance of works consists of the following parts:

- Part 1: Works on Tracks
- Part 2: Works on switches
- Part 3: Acceptance of rail grinding, milling and planing work in track

1 Scope

This part of this European Standard lays down the technical requirements and the measurements to be made for the acceptance of work to re-profile the heads of railway rails by grinding, milling or planing, including switches and crossings. It also specifies the procedures to be followed both to verify reference instruments to be used for these measurements and to approve non-reference instruments to be used for measurements.

It does not apply to tramway rails, nor to rails of less than 40kg/m nominal linear mass. It is also not applicable to welds when these are re-profiled as part of their production process.

2 Normative references

This part of this European Standard incorporates by dated or undated reference, provisions from other publications. These normative references are cited at appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this European Standard only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies.

ISO 3274:1988, *Geometrical Product Specifications (GPS) - Surface texture: Profile method - Nominal characteristics of contact (stylus) instruments*

ISO 4288:1996, *Geometrical Product Specifications (GPS) - Surface texture: Profile method - Rules and procedures for the assessment of surface texture*

DIN 4768:1990, *Determination of values of surface roughness parameters R_a , R_z , R_{max} using electrical contact (stylus) instruments; concepts and measuring conditions*

ISO 11562:1998, *Geometrical Product Specifications (GPS) - Surface texture: Profile method - Metrological characteristics of phase correct filters*

ISO/FDIS 14253-1 (Final Draft, 1998), *Geometrical Product Specifications (GPS) - Inspection by measurement of workpieces and measuring equipment - Part 1: Decision rules for proving conformance or non-conformance with specification*

3 Definitions

For the purposes of this Standard, the following definitions apply:

3.1 reference profile

the transverse profile to which rail is to be ground, within the specified tolerances

3.2 reference line

the line normal to the track's longitudinal axis and tangent to the heads of both rails

3.3 angle of inclination of rail

the nominal angle at which rail is laid (Fig 1(b)) e.g. vertical, 1:20, 1:30, 1:40 (equivalent to 0, 2.86, 1.91 and 1.43 degrees respectively)

NOTE - For rail which is laid in non-canted track, the angle of inclination of the rail is equal to the angle between the vertical and the centreline of the inclined rail.

3.4 angle of inclination of profile

the angle at which the reference profile is to be ground on the rail (Fig 1(a))

NOTE - Usually the angle of inclination of the profile is equal to the desired angle of inclination of the rail but the two may differ. For example, in the case of switch and crossing work, it is common to lay rails vertically i.e. with zero inclination. On these rails, however, a profile may be ground at an inclination which matches that of the adjacent plain line e.g. 1 in 20.

3.5 reference rail

a rail with the desired reference profile, laid at the specified angle of inclination of the profile;

NOTE - The angle of inclination of the reference rail is usually equal to the angle of inclination of the reference profile, but this is not so in all cases (Figs 1(b) and (c)).

3.6 reference point A

That point towards the gauge side of a reference rail at which the angle between the reference line and the tangent to the profile is equal to the specified angle of inclination of the profile (Fig 1(a)).

NOTE - For typical, symmetric reference profiles, reference point A is identical to the point at which a line through the centre of the web of the reference rail intersects the reference profile (Fig 1(b)).

3.7 reference point B

That point on the gauge face of a reference rail which lies 14mm below that line which is parallel to the reference line and which passes through reference point A (Fig 1(a)).

3.8 reference point B'

That point on the gauge corner of a reference rail at which a line which is tangent to the rail lies at an angle of 45 degrees to the reference line (Fig 1(d)).

3.9 reprofiling zone

the area of the head of a reference rail between angles of 70 degrees towards the gauge corner and 5 degrees towards the field corner from the line normal to the reference line (Fig 2)

3.10 reference instrument

an instrument for the measurement of longitudinal or transverse profile the performance of which has been verified in accordance with Annex A of this Standard.

3.11 approved instrument

an instrument for measurement of longitudinal or transverse profile the usage of which is justified by correlation of its performance with that of a reference instrument in accordance with Annex B of this Standard

3.12 test instrument

an instrument whose use as a reference instrument or an approved instrument is being tested

3.13 grinding site

a continuous length of track where the rail is to be ground

3.14 grinding facet

an approximately plane sector of the profile of the ground rail which is produced by a single grinding stone whose axis of rotation is normal to the rail's longitudinal axis

3.15 characteristic length

the length on the rail travelled during one rotation of a grinding stone whose axis of rotation is normal to the rail's longitudinal axis

3.16 traced profile¹

locus of the tip of the measuring instrument as it traverses the rail surface

NOTE - This is the profile from which all other profiles used in this Standard are derived.

3.17 profile filter²

an electronic device which separates profiles into longwave and shortwave components, or into components within a specified wavelength range

3.18 phase correct profile filter³

profile filter which does not cause phase shifts

3.19 cut-off wavelength⁴

wavelength of a sinusoidal profile of which 50% of the amplitude is transmitted by the profile filter

NOTE - Profile filters are identified by their cut-off wavelength value

3.20 filtered profile

the profile which results from applying a profile filter to the primary longitudinal profile

3.21 moving average of peak-to-peak amplitudes

the average depth of individual irregularities in a profile which is calculated as a quasi-continuous function of distance along the track; the depth is calculated for that section of the profile which lies within a "window" of specified length (Fig 3); the function with distance is calculated by incrementing the window along the profile by a distance equal to the sampling interval.

¹ This definition is similar to that for instruments to measure surface texture, as given in ISO 3274.

² This definition is similar to that for instruments to measure surface texture, as given in ISO 11562:1998.

³ This definition is similar to that for instruments to measure surface texture, as given in ISO 11562:1998.

⁴ This definition is similar to that for instruments to measure surface texture, as given in ISO 11562:1998.

3.22 moving average of root-mean-square (RMS) amplitudes

the root-mean-square amplitude of a profile which is calculated as a quasi-continuous function of distance along the track; the average is calculated for that section of the profile which lies within a "window" of specified length (Fig 4); the function with distance is calculated by incrementing the window along the profile by a distance equal to the sampling interval

3.23 transition length

initial or final section of a length of track where the validity of a measurement of longitudinal or transverse profile is questionable for a variety of reasons, including settling of electronic components and circuits

3.24 sampling interval

spacing along the track at which a continuous record of the traced profile is sampled in order to produce the primary profile

3.25 primary profile⁵

digital representation of the traced profile before application of any profile filter

3.26 percentage exceedence

the percentage length of a test site over which a measurement of the amplitude of the filtered profile exceeds a prescribed limit

4 Longitudinal profile

4.1 principle

Measurements are made using either a precision system, a "reference instrument", or an "approved" instrument. Approved instruments do not offer the same accuracy as reference instruments but are generally adequate for the purpose of demonstrating compliance with the requirements of this standard. An example is the type of system used for routine measurements on grinding trains.

In accordance with historic practice, limits are set on the magnitude of the irregularities that can remain in track after a reprofiling operation. It is recognised however that it can be uneconomic to

⁵ This definition is a working form of the definition of a "primary profile" given in ISO 3274 for measurement of surface roughness, which is more appropriate to the requirements of this standard.

achieve 100% compliance with these, particularly where isolated top faults, such as wheelburn, exist prior to reprofiling. Two standards are therefore offered, differentiated by the percentage of the reprofiled track meeting the specified criteria. Where isolated top faults exist, Class 2 offer a lower cost option compared to Class 1 as it will be achieved with fewer passes. However a larger number of isolated non-compliant zones will remain in the reprofiled site.

Class 1 also includes limits for very short (10-30mm) and very long (300-1000mm) wavelength residual irregularities; these are not included in Class 2. Where corrugations in these wavebands are required to be removed it will also be necessary to specify Class 1.

This standard permits the quality of the longitudinal profile to be characterised in any one of the three ways, at the discretion of the client. These are the moving average RMS amplitude of the irregularities in a given waveband, the moving average peak-to-peak amplitude of irregularities in a given waveband and the number of irregularities exceeding the specified limit per 100m of track. Except in borderline cases all three methods are expected to give similar results.

4.2 measurements required

The longitudinal profile of the finished ground rail shall be recorded continuously using either a reference instrument or an approved instrument.⁶ In cases of dispute a reference instrument shall be used. All measurements undertaken in order to demonstrate compliance with Section 4.3 of this standard shall be recorded.

Measurements shall be undertaken at the latest within 8 days of grinding or after the track has carried 0.3MGT of traffic, whichever occurs sooner.

NOTE - It is preferable for measurements to be made immediately after grinding.

Longitudinal profile measurements shall be made within a distance of 15mm laterally on the rail from reference point A, to produce the traced profile.

NOTE - It is recommended that a digital form of the traced profile, the primary profile, is used for subsequent analysis.

⁶ Procedures are set out in Annexes A and B of this Standard respectively for verification of instruments and to demonstrate correlation between approved and reference instruments

4.3 acceptance criteria

The rail's longitudinal profile shall be ground to one of two Classes as stated in the Contract.

If Class 1 is required, the primary or traced profile shall be processed to provide a filtered profile within each of the four wavelength ranges given in Table 1. If instead Class 2 is required, the primary or traced profile shall be processed to provide filtered profiles in the 30-100mm and 100-300mm wavelength ranges.

Conformance with the Standard shall be demonstrated by calculating one of the following three quantities, as specified in the Contract:

- the Moving Average of root-mean-square (RMS) amplitudes of individual irregularities in the filtered profiles
- the Moving Average of peak-to-peak amplitudes of individual irregularities in the filtered profiles,
- the number of irregularities in each filtered profile whose peak-to-peak amplitude exceeds the limits specified in line 1 of Table 2.

The cut-off wavelengths for each wavelength range and the length of the corresponding window within which the pertinent Moving Average is to be calculated are stated in lines 0 and 1 of respectively of Table 1 e.g. for the 30-100mm wavelength range, the cut-off wavelengths are 30mm and 100mm, and the window length is 0.6m.

NOTE - For measurements in the 10-30mm wavelength range, it is at present unlikely that instruments other than reference instruments will have sufficient accuracy.

If the Moving Average of root-mean-square (RMS) amplitudes is specified in the Contract as the quantity to be used to assess acceptability of the ground rail, this Moving Average shall exceed the limit shown in line 2 of Table 1 for no more than the percentage of the length of the grinding site shown in line 4 of Table 1 if Class 1 is specified. If Class 2 is specified, this Moving Average in the wavelength ranges 30-100mm and 100-300mm shall exceed the limit shown in line 2 of Table 1 for no more than the percentage of the length of the grinding site shown in line 5 of Table 1.

If the Moving Average of peak-to-peak amplitudes is specified in the Contract as the quantity to be used to assess acceptability of the ground rail, this Moving Average shall exceed the limit shown in line 3 of Table 1 for no more than the percentage of the length of the grinding site shown in

line 4 of Table 1 if Class 1 is specified. If Class 2 is specified, this Moving Average in the wavelength ranges 30-100mm and 100-300mm shall exceed the limit shown in line 3 of Table 1 for no more than the percentage of the length of the grinding site shown in line 5 of Table 1. If the number of irregularities exceeding a specified amplitude is used as the criterion to assess acceptability of the ground rail, this number for any 100m length of the ground rail for each of the wavelength ranges shown in line 1 of Table 2 shall not exceed the limit shown in line 2 of Table 2 if Class 1 is specified. If Class 2 is specified, this number for any 100m length of the ground rail for the wavelength ranges 30-100mm and 100-300mm shall not exceed the limit shown in line 3 of Table 2

NOTE - The limits on allowable RMS and peak-to-peak amplitudes are illustrated in Figures 5 as functions of the wavelength.

Table 1 Acceptance criteria for longitudinal profile of ground track expressed in terms of Moving Averages

line					
0	wavelength range (mm)	10-30	30-100	100-300	300-1000
1	window length (m)	0.6	0.6	1	5
2	limit on moving average of RMS amplitudes (mm)	0.004	0.004	0.012	0.040
3	limit on moving average of peak-to-peak amplitudes (mm)	0.010	0.010	0.030	0.100
4	Class 1, allowable percent exceedence of limit (%)	5	5	5	10
5	Class 2, allowable percent exceedence of limit (%)	-	10	10	-

NOTE - In switch and crossing work, not all rail can be ground

Table 2 Acceptance criteria for longitudinal profile of ground track expressed in terms of numbers of exceedences

line	wavelength range (mm)	10-30	30-100	100-300	300-1000
1	limit on peak-to-peak amplitudes (mm)	0.010	0.010	0.030	0.100
2	Class 1, allowable number of individual irregularities exceeding limit per 100m of ground rail	250	100	25	10
3	Class 2, allowable number of individual irregularities exceeding limit per 100m of ground rail	-	200	50	-

NOTE - In switch and crossing work, not all rail can be ground

5 Transverse profile

5.1 principle

Measurements are made using either a precision, a "reference instrument", or an "approved" instrument.⁷ Approved instruments do not offer the same accuracy as reference instruments but are generally adequate for the purpose of demonstrating compliance with the requirements of this standard. Some of the systems used for routine measurements on grinding trains fall into this category.

Reprofiling may be undertaken for a variety of reasons. Where reprofiling is undertaken purely for the removal of corrugation, there may be little need for the transverse profile to be ground with precision. Where the objective is to control, however, it is necessary for the ground profile to match closely the ideal, represented by the "reference rail". A range of Classes is therefore included to enable the client to specify the level of precision which is appropriate for the site to be reprofiled.

⁷ Procedures are set out in Annexes A and B of this Standard respectively for verification of measuring instruments and to demonstrate correlation between approved and verified reference instruments.

The match between the as-ground profile and ideal profile (i.e. the profile of the reference rail) is determined by aligning the two at two points and measuring maximum difference between them (see Fig 6). For straight track, these points of alignment generally approximate to the rail crown and the gauge point. On the high rail of curves this method is not applicable if significant sidewear has occurred and an alternative method of alignment is therefore used.

5.2 measurements required

The rail's transverse profile shall be measured using either a reference instrument or an approved instrument. In cases of dispute a reference instrument shall be used. All measurements undertaken in order to demonstrate compliance with Section 5.3 of this Standard shall be recorded.

Measurements shall be made at the latest within 8 days of grinding or after the track has carried 0.3MGT of traffic, whichever occurs sooner.

NOTE - It is preferable for measurements to be made immediately after grinding.

The transverse profile of each finished, ground rail shall be measured sufficiently frequently to ensure compliance with the requirements stated in Section 5.3 of this Standard. The transverse profile shall be recorded at least once per grinding site or at an interval of less than or equal to 500m on a grinding site greater than 500m long.

In the case of a dispute regarding acceptance of the transverse profile of the ground rail, measurements of each rail shall be made at an interval of greater than or equal to 10m throughout the grinding site.

5.3 acceptance criteria

Acceptance of the transverse profile of the ground rail is based only on a comparison of the measured profile and the reference profile in the reprofiling zone defined in Section 3 of this Standard.

Each measured profile shall be aligned with the profile of the appropriate reference rail so that the profiles coincide at two points. The alignment shall be undertaken without rotation of either profile. For tangent track and for the low rail in curves, these two points shall be reference points

A and B, as defined in Section 3 of this Standard. For the high rail in curves, these two points shall be reference points A and B', as defined in Section 3 of this Standard.

NOTE - For rail which is side worn at other sites e.g. where there is hunting in tangent track, it is recommended that the alignment is undertaken at reference points A and B'.

The deviation of each measured profile from the profile of the reference rail shall be calculated normal to the tangent to the reference rail (Fig 6). Between the two points at which the profiles are aligned, the maximum positive and negative deviations shall be found. A deviation is positive if the measured profile lies above the reference profile.

The rail's transverse profile shall be ground to one of three Classes, as shown in Table 3. The Class to be achieved shall be stated in the Contract. For each Class, at least the tabulated percentages of individual measurements of the deviation of the transverse profile from the profile of the reference rail shall lie within the tabulated limits e.g. for rail ground to Class Q, 100% of individual measurements shall deviate within a range of 1.0mm, and at least 80% within a range of 0.6mm.

The interpretation of the range of deviation shall be stated in the Contract. A range of deviation of 0.6mm shall be interpreted as allowing +/-0.3mm deviation from the reference profile, or +0.2mm / -0.4mm; or +0mm / -0.6mm etc.

Measurements shall be undertaken and deviations of measurements of the ground rail from the reference rail shall be calculated as described in Section 5.1 of this Standard.

Table 3 Acceptance criteria for transverse profile of ground track

	range of deviation (mm)		
Class	0.6	1.0	1.7
Q	80	100	100
R	-	80	100
S	-	-	80

NOTES

- In switch and crossing work, not all rail can be ground
- Where reprofiling is undertaken to control conicity, Class Q is likely to be appropriate.

6 Metal removal

6.1 measurements required

Measurements of metal removal from the railhead are required only if there is a requirement in the Contract to demonstrate a minimum or maximum depth of metal removal. All measurements undertaken in order to demonstrate compliance with Section 6.2 of this Standard shall be recorded.

The height of the rail shall be measured using a micrometer whose accuracy is in accordance with BS**** / ISO ***⁸

The rail depth or depth of the railhead shall be measured before and after grinding at a minimum of 5 positions on each rail a distance of no less than 0.5m apart. Measurements shall be made within a month of grinding. The rail shall be marked to ensure that measurements before and after grinding are made within a distance of 10mm of each other along the rail. If the rail is initially corrugated, measurements shall be undertaken in the trough of the corrugation.

Measurement of the rail height or depth of the railhead shall be processed so as to provide the depth of metal removed within 15mm transversely of reference point A, or elsewhere on the railhead as agreed between Client and Contractor.

Measurements shall be recorded once per week or as required by the Contract.

6.2 acceptance criterion

The minimum or maximum depth of metal removed shall be as required by the Contract at the minimum percentage of measurement positions specified by the Contract.

⁸ Procedures are set out in Annexes A and B of this Standard respectively for verification of measuring instruments and to demonstrate correlation between approved and verified reference instruments.

7 Surface roughness

7.1 measurements required

Measurements of the surface roughness of the finished, ground rail shall be made immediately after grinding, within 10mm of reference point A defined in Section 3 of this Standard, on the same grinding facet and along the rail's longitudinal axis. At least six contiguous measurements shall be taken to cover at least one complete characteristic length on the grinding facet, as defined in Section 3 of this Standard. The arithmetic mean surface roughness shall be recorded for each of the measurements.

Measurements shall be recorded once per week or as required by the Contract.

In cases of dispute, roughness shall be measured using an instrument which complies with the requirements of ISO 3274:1998.⁹ Measurements shall be made with a roughness sampling length $l_r = 2.5mm$ and a roughness evaluation length $l_n = 12.5mm$, as described in ISO 4288:1996.

7.2 acceptance criteria

The ground rail shall be considered acceptable if no more than one of the 6 values of arithmetical mean roughness, R_a , or 16% of the total number of values, exceeds 10 microns.

NOTE - Other values may apply e.g. for reasons of noise abatement or improved adhesion.

8 Visual appearance: acceptance criteria

The maximum facet width shall be 4mm on the gauge corner, 7mm on the shoulder and 10mm on the rail crown. The reprofiling zone shall be blended smoothly into the parent rail.

The maximum variation in facet width over a 100mm length of rail shall be 25% of the maximum width of the facet.

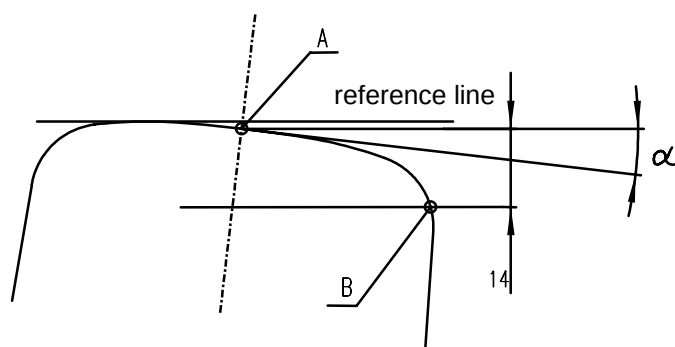
There shall not be continuous blueing in the reprofiling zone.

⁹ Verification of the performance of the test instrument is undertaken by comparing RMS amplitudes of filtered profiles. It is accordingly unnecessary for the filtering here to be undertaken using phase correct profile filters

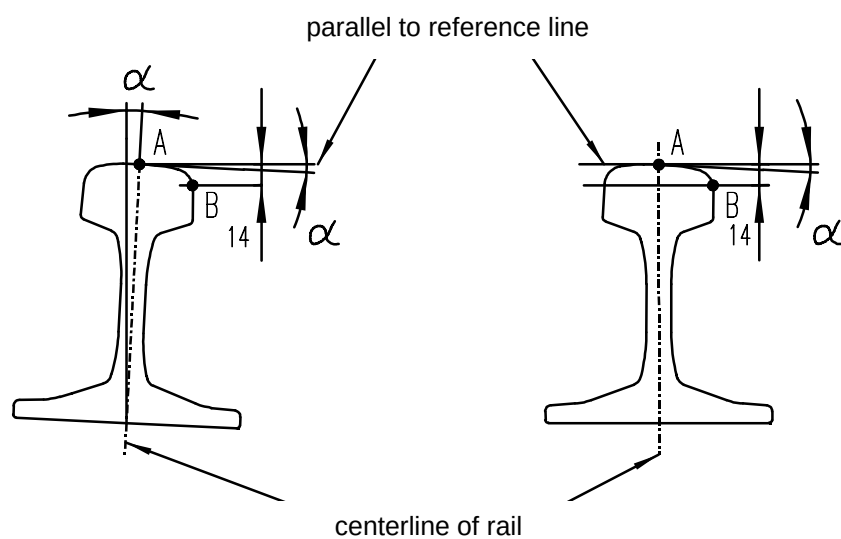
9 Acceptance documentation

The form of acceptance documentation required shall be stated in the Contract.

NOTE - An example of acceptance documentation is given in Annex C.

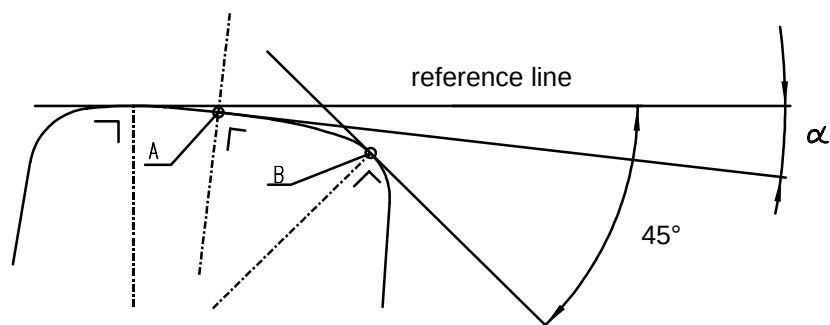


(a) Detailed section of railhead, showing reference points A and B (α :angle of rail inclination)



(b) Illustration on inclined rail with symmetric reference profile ground at the same angle of inclination.
(α :angle of rail inclination)

(c) Illustration on non-inclined rail with asymmetric reference profile ground at specified angle of inclination.
(α :angle of rail inclination)



(d) Detailed section of railhead, showing reference point A and B'. (α :angle of rail inclination)

Figure 1 : Definition of terms, and determination of reference points A, B and B' on the transverse profile.

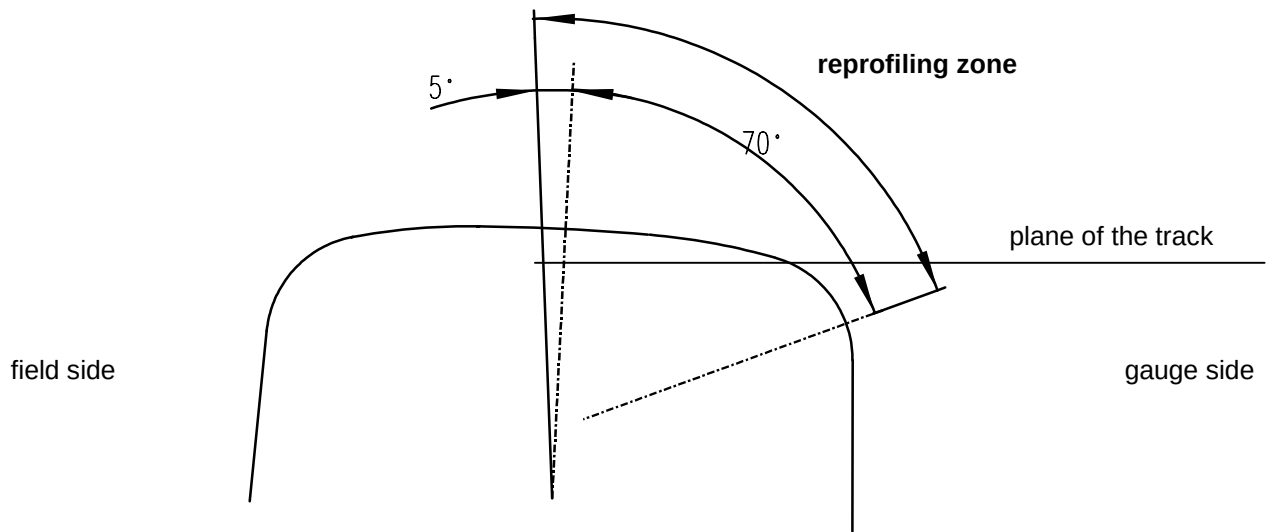
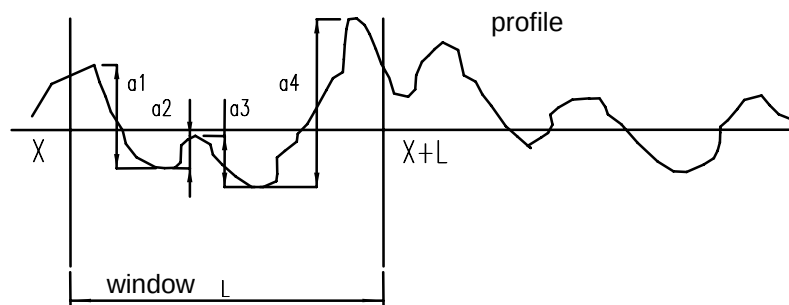


Figure 2 : Reprofile zone



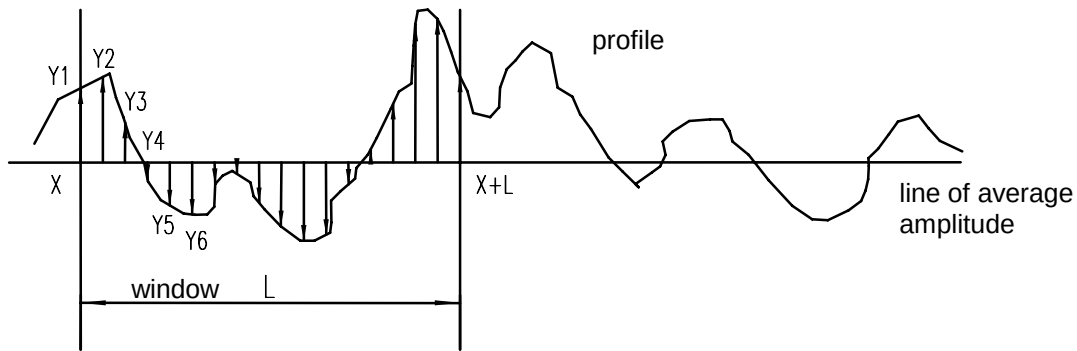
Moving average of peak-to-peak amplitudes within the window of length L

$$PP(x,L) = (a_1 + a_2 + \dots + a_n)/n$$

where there are n individual irregularities within the window of length L.e.
In the specific case shown,

$$PP(x,L) = (a_1 + a_2 + a_3 + a_4)/4$$

Figure 3 : Moving average of peak-to peak amplitude of a profile



The moving average of root-mean square (RMS) amplitudes of the profile $y(x)$ in a window of length L is in general

$$RMS(x, L) = \left(\frac{1}{L} \int_x^{x+L} y^2(x) dx \right)^{1/2}$$

If a digitized representation of the profile is used, with values $y(t)$, $y(t+\Delta t)$, $y(t+2\Delta t)$ etc. then equivalent RMS amplitude of the digitized profile is

$$RMS(x, L) = \left(\sum_{i=1}^{i=n} y_i^2 / (n-1) \right)^{1/2}$$

where there are n samples of the profile y within the window of length L .

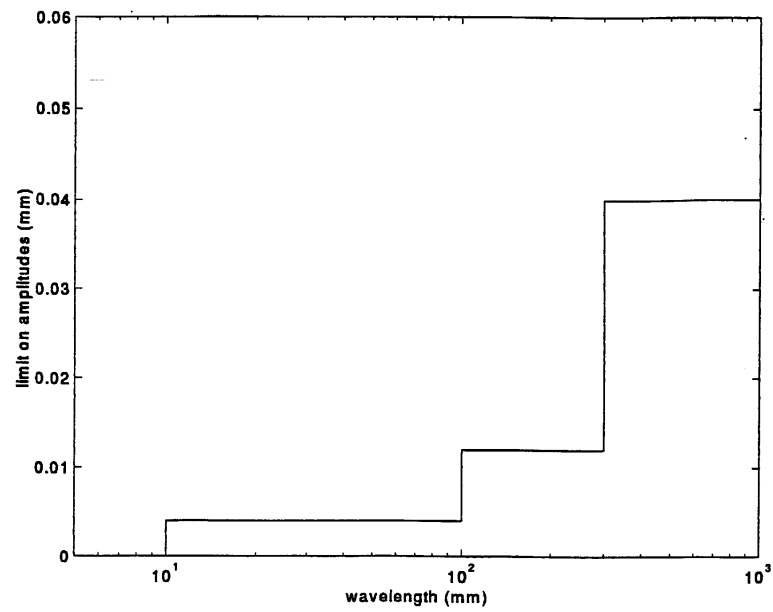
The RMS amplitude of the traced profile in a specified wavelength range (λ_1, λ_2) can be calculated as shown above by first filtering the traced profile to give the filtered profile.

Alternatively the Power Spectral Density $S(x, L, \lambda)$ of the traced profile within the window of length L can first be calculated. The RMS amplitude in the specified wavelength range is then.

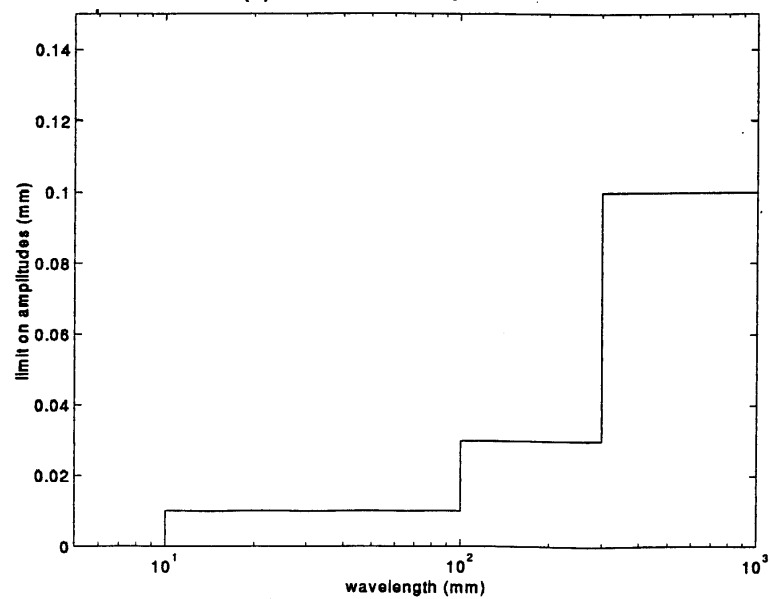
$$RMS(x, L, \lambda_1, \lambda_2) = \left(\int_{\lambda_2}^{\lambda_1} S(x, L, \lambda) d\lambda \right)^{1/2}$$

Calculation of the Power Spectral Density of a digitized signal is a standard signal processing function.

Figure 4 Moving average of root-mean-square amplitudes of a profile



(a) limit on RMS amplitudes



(b) limit on peak-to-peak amplitudes

Figure 5 Limits on RMS and peak-to-peak amplitudes of the longitudinal profile of a ground rail as functions of wavelength

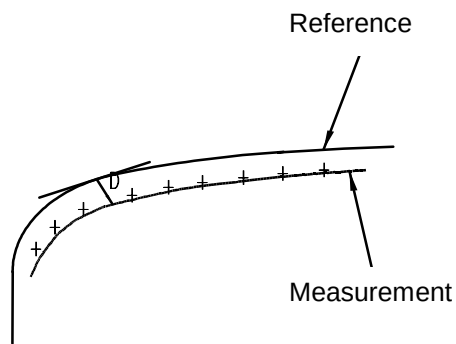


Figure 6 : Deviation of measured transverse profile from profile reference

Annex A **(normative)** **Procedures to verify reference instruments**

A.1 Longitudinal profile

A.1.1 principle

The longitudinal profile of a line along a machined strip on a purpose-built beam is measured using both the test instrument and a Coordinate Measuring Machine (CMM), the accuracy of which has been verified. The test instrument is deemed to be verified as a reference instrument for measurement of the longitudinal rail profile if there is satisfactory correlation between root-mean-square (RMS) amplitudes of a number of filtered profile measurements made using the test instrument and the CMM.

NOTE - An example of a calibration beam suitable for verification of instruments for wavelengths of less than 300mm is shown in Fig A.1.

A.1.2 calibration beam

Requirements of the machined strip on the calibration beam are as follows:

- the machined strip shall be at least 20mm wide throughout its length
- the machined strip shall be at least three times as long as the maximum wavelength of the range for which the test instrument is to be verified; these lengths are given in line 1 of Table A.1
- for verification of a test instrument to measure wavelengths of less than 1000mm, there shall exist irregularities in the track whose lengths are in the range 10-1000mm and whose depths are in the ranges 0.0-0.01mm and 0.0-0.5mm

NOTES - The longitudinal profile of the nominal centre line of the machined strip on the calibration beam shown in Fig A.1 is shown in Fig A.2.

- The longer of the longitudinal irregularities on a beam such as that in Fig A.1 can be machined by traversing the machined strip under a grinding head, and incrementing the grinding head up and down. The shorter irregularities can be produced by finishing.

- Because the line along the machined strip is measured on the assembled calibration beam, the actual form of the irregularities which are machined is not of critical importance.

**Table A.1 Parameters for calibration beam for verification of reference instrument to
measure longitudinal rail profile**

line	wavelength range (mm)	10-30	30-100	100-300	300-1000
1	minimum length (mm) of machined strip on calibration beam	90	300	900	3000

A.1.3 Coordinate Measuring Machine

Requirements of the Coordinate Measuring Machine are as follows:

- The continuous measuring length shall be no less than three times as long as the maximum wavelength of the range for which the test instrument is to be verified e.g. if the test instrument is to be verified for measurement of wavelengths in the range 100-300mm, the continuous measuring length shall be at least 900mm long.
- A valid Test Certificate issued by a certified Testing Authority shall be available. The linear accuracies of the Coordinate Measuring Machine along x, y and z axes and the method by which these have been found shall be recorded. These linear accuracies shall be better than 0.001mm.

A standard, general-purpose measuring probe with a head diameter in the range 3-6mm shall be used for all measurements. The head diameter, the orientation of the calibration beam relative to the x, y and z axes, and the temperature at which measurements are taken shall be recorded. At least three measurements of the longitudinal profile of the calibration beam shall be recorded in each direction along nominally the same track on the beam. The calibration beam shall be held in such a way as to minimise irregularities in the beam which may arise from the holding arrangement itself.

A primary profile shall be produced for each of the traced profiles. The sampling interval along the calibration beam and the digitisation increment for this primary profile shall be recorded. The sampling interval shall be less than or equal to 2mm and the digitisation increment shall be less than or equal to 1 micron. A filtered profile shall be produced for each of the wavelength ranges stated in Table A.2 for which the test instrument is to be verified. The filters shall attenuate wavelength components in the primary profile at a rate of at least 60dB/decade outside the wavelength range of interest (i.e. the filters shall be at least 6th order).

NOTE - Examples of satisfactory filter characteristics are shown in Fig A.3.⁹

The root-mean-square (RMS) amplitude of the filtered profile shall be calculated by summation e.g. if there are n individual samples z_i of the filtered profile within the measuring length, the RMS amplitude is¹⁰

$$z_{RMS} = \left[\sum_{i=1}^{i=n} z_i^2 / (n-1) \right]^{1/2} \quad (A\ 1)$$

The repeatability of the CMM shall be considered acceptable for a particular wavelength range if the maximum difference between RMS amplitudes of any pair of CMM measurements is less than the relevant value shown in line 2 of Table A.2.

Table A.2 Requirements of the Coordinate Measuring Machine for verification of reference instrument to measure longitudinal rail profile

line	wavelength range (mm)	10-30	30-100	100-300	300-1000
1	minimum continuous measuring length (mm)	90	300	900	3000
2	maximum difference between any pair of RMS amplitudes (mm) from CMM records	0.001	0.001	0.001	0.002

⁹ Verification of the performance of the test instrument is undertaken by comparing RMS amplitudes of filtered profiles. It is accordingly unnecessary for the filtering here to be undertaken using phase correct profile filters

¹⁰ Verification of the performance of the test instrument is undertaken by comparing RMS amplitudes of filtered profiles. It is accordingly unnecessary for the filtering here to be undertaken using phase correct profile filters

A.1.4 test instrument

At least three measurements of the longitudinal profile of the calibration beam shall be recorded in each direction along nominally the same line on the calibration beam using the test instrument. The temperature at which these measurements are taken shall be recorded. During the measurements, the calibration beam shall be held in such a way as to minimise irregularities in the beam which may arise from the holding arrangement itself.

In order to minimise errors arising from temperature variations and from transport of the test instrument and calibration beam, measurements using this instrument and the CMM shall be undertaken in the same location and on the same occasion. The beam shall be supported in such a way as to minimise errors arising from deflection under load.

RMS amplitudes of the filtered profiles for each of the traced profiles shall be calculated for each of the wavelength ranges stated in Table A.3 for which the test instrument is to be verified. These calculations are to be undertaken using the same algorithms and filter characteristics as described in Section A.1.3 for measurements made using the CMM.

The overall RMS amplitude of the RMS amplitudes for each of the six sets of filtered profiles for the CMM and for the test instrument shall be calculated for each of the wavelength ranges for which the test instrument is to be verified. The overall RMS amplitude for a set of six filtered profiles is

$$Z = \left[\sum_{j=1}^{j=6} n_j z_j^2 / \left(\sum_{j=1}^{j=6} n_j - 1 \right) \right]^{1/2} \quad (A.2)$$

where z_j is the RMS amplitude and n_j is the number of samples for the j th set of filtered profiles.

The repeatability of the test instrument shall be considered acceptable for a particular wavelength range if the maximum difference between RMS amplitudes of any pair of measurements made using this instrument is less than the relevant value shown in line 2 of Table A.3.

The test instrument shall be verified as a reference instrument for a particular wavelength range if the maximum difference between the overall RMS amplitude for the CMM and for the test instrument (both calculated from eqn (A.2)) is less than the value shown in line 3 of Table A.3 for that wavelength range.

NOTE - For reference instruments which are to be used to determine the acceptability of acoustic grinding, these verification criteria may be inadequate and it may be necessary to compare amplitudes over narrower wavelength ranges.

Table A.3 Criteria for verification of reference instrument to measure longitudinal rail profile

line	wavelength range (mm)	10-30	30-100	100-300	300-1000
1	minimum lengths to be measured on calibration beam (mm)	90	300	900	3000
2	maximum difference between any pair of RMS amplitudes (mm) from records of test instrument	0.001	0.001	0.002	0.005
3	maximum difference between overall RMS amplitudes (mm) from CMM and test instrument	0.002	0.002	0.004	0.010

A.1.5 documentation

In order for an instrument to be used as a reference instrument for the purposes of this Standard for measurement of the longitudinal rail profile, the above results shall be documented and the primary profiles for each measurement shall be available for inspection by a Client.

A statement shall be made of the following:

- the wavelength range(s) for which the instrument has been verified
- the temperature (or temperatures) at which measurements were undertaken for purposes of verification
- the temperature range over which the supplier of the test instrument believes that the verification is valid, if this is greater than the temperature range at which measurements were undertaken
- the reason(s) why the validation is acceptable for this temperature range.

A.2 Transverse profile

A.2.1 principle

Two rail sections are mounted in a calibration jig which holds the sections at known inclination and at a separation within the range of gauges for which a reference instrument would be used. The profile of one rail is close to a reference profile, while that of the other differs significantly from the reference profile. The transverse profile of nominally the same line across the two rail sections is measured using both the test instrument and a Coordinate Measuring Machine (CMM), the accuracy of which has been verified. The test instrument is deemed to be verified as a reference instrument for measurement of the transverse rail profile if there is satisfactory correlation between measurements made using the test instrument and the CMM.

A.2.2 calibration jig

An example of a calibration jig is shown in Fig A.4. Requirements of the calibration jig are as follows:

- two rail sections are rigidly mounted in the jig so that they are parallel to one another, at nominally the same angle of inclination and at a nominal separation in the range 1-2m
- the rail sections shall be of identical height
- the rail sections shall be held sufficiently rigidly that they do not deflect significantly during measurement
- the reference profile and angle of inclination of the profile shall be stated
- the transverse profile of one rail shall be such that it deviates throughout the reprofiling zone by less than $\pm 0.3\text{mm}$ from the profile of the reference rail
- the transverse profile of the other rail shall be such that it deviates by at least 0.5mm from the profile of the reference rail at one or more points in the reprofiling zone

NOTES

- It is recommended that the reference rail be one with a common transverse profile and at a common angle of inclination e.g. UIC60 at an inclination of 1:20.
- It is recommended that a separation of 1435mm, equivalent to standard gauge track, be used.

To demonstrate acceptability of the test jig, each of the six profile measurements made as described in Annex A.2.3 shall be aligned at reference points A and B on the profile of the reference rail. The difference between the measured profile and the profile of the reference rail

shall be calculated at increments of no more than 2.5 degrees or 1mm, whichever is the smaller, around the reference rail within the reprofiling zone i.e. from 5 degrees to field to 70 degrees to the gauge side of the rail. These measurements shall demonstrate that the profile of one rail section differs by less than $\pm 0.3\text{mm}$ from the profile of the reference rail and that the profile of the second rail section differs by more than $\pm 0.5\text{mm}$ from the profile of the reference rail.

A.2.3 Coordinate Measuring Machine

Requirements of the Coordinate Measuring Machine are as follows:

- It shall be sufficiently large as to measure the transverse profiles of the two rail sections in the calibration jig without moving the jig.
- A valid Test Certificate issued by a certified Testing Authority shall be available. The linear accuracies of the Coordinate Measuring Machine along x, y and z axes and the method by which these have been found shall be recorded. These linear accuracies shall be better than 0.001mm.

A standard, general purpose measuring probe with a head diameter in the range 3-6mm shall be used for all measurements. The head diameter, the orientation of the calibration jig relative to the x, y and z axes, and the temperature at which measurements are taken shall be recorded. At least three measurements of the transverse profile of both rail sections shall be recorded.

A primary profile shall be produced for each of the traced profiles. The sampling interval around the rail section and the digitisation increment for the primary profile shall be recorded.

To demonstrate satisfactory repeatability of the CMM, each one of the three measurements of the two rail sections made using the CMM shall be aligned with the other two measurements. For this purpose, each measured profile shall be taken in turn as if it were the reference profile on a reference rail. Reference points A and B on this profile shall be found as defined in Section 3 of this Standard for the specified angle of inclination of the profile. The other two profile measurements shall be aligned in turn so that each coincides at reference points A and B on the first measured profile. The difference between the profiles shall be calculated at increments of no more than 2.5 degrees around the reference rail within the reprofiling zone i.e. from 5 degrees to field to 70 degrees to the gauge side of the rail. The maximum and minimum difference between each pair of profiles shall be recorded.

NOTE - This requires a total of 6 comparisons to be made for each of the two rail sections. If there are three measurements A, B and C, these shall be permuted AB, AC, BA, BC, CA, CC.

- The maximum and minimum differences between the pairs of measurements can conveniently be tabulated as shown in Table A.4.

The repeatability of the CMM shall be considered acceptable if the maximum difference between any pair of the three CMM measurements of each of the two rail sections is less than 0.005mm.

A.2.4 test instrument

At least three measurements of the transverse profile of both rail sections shall be recorded using the test instrument. Between each of the three measurements using the test instrument, the instrument shall be removed and replaced on the rail.

A primary profile shall be produced for each of the traced profiles. The sampling interval around the rail section and the digitisation increment for the primary profile shall be recorded.

To demonstrate satisfactory repeatability of the test instrument, each one of the three measurements of the two rail sections made using the test instrument shall be aligned with the other two measurements. For this purpose, each measured profile shall be taken in turn as if it were the reference profile on a reference rail; reference points A and B on this profile shall be found as defined in Section 3 of this Standard for the specified angle of inclination of the profile; and the other two profile measurements shall be aligned in turn so that each coincides at reference points A and B on the first measured profile. The difference between the profiles shall be calculated at increments of no more than 2.5 degrees around the reference rail within the reprofiling zone i.e. from 5 degrees to field to 70 degrees to the gauge side of the rail. The maximum and minimum difference between each pair of profiles shall be recorded.

NOTE - This requires a total of 6 comparisons to be made for each of the two rail sections. If there are three measurements X, Y and Z, these shall be permuted XY, XZ, YX, YZ, ZX, ZY.

- The maximum and minimum differences between the pairs of measurements can conveniently be tabulated as shown in Table A.4.

The repeatability of the test instrument shall be considered acceptable if the maximum difference between any pair of the three measurements of each of the two rail sections is less than 0.050mm.

To demonstrate satisfactory correlation of measurements made with the test instrument and with the CMM, each one of the three measurements of the two rail sections made using the test instrument shall be aligned with each of the three measurements of the rail section made with the CMM. For this purpose, each profile measured with the CMM shall be taken in turn as if it were the reference profile on a reference rail; reference points A and B on this profile shall be found as defined in Section 3 of this Standard for the specified angle of inclination of the profile; and the profile measurement for the test instrument shall be aligned in turn so that it coincides at reference points A and B on the CMM profile. The difference between the profiles shall be calculated at increments of no more than 2.5 degrees around the reference rail within the reprofiling zone i.e. from 5 degrees to field to 70 degrees to the gauge side of the rail. The maximum and minimum difference between each pair of profiles shall be recorded.

NOTE - This requires a total of 9 comparisons to be made for each of the two rail sections. If there are three measurements A, B, C for the CMM and X, Y, Z for the test instrument, these shall be permuted AX, AY, AZ, BX, BY, BZ, CX, CY, CZ.

- The maximum and minimum differences between the pairs of measurements can conveniently be tabulated as shown in Table A.5.

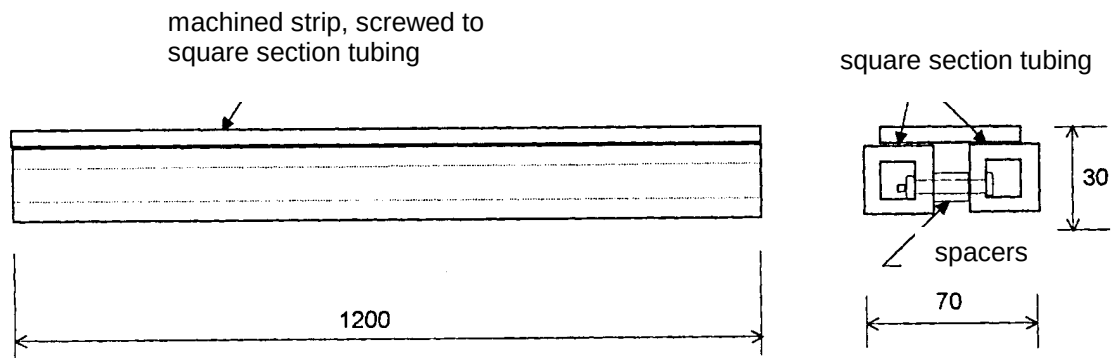
The test instrument shall be verified as a reference instrument if the maximum difference between any pair of measurements made with the CMM and with the test instrument is less than 0.100mm.

A.2.5 documentation

In order for an instrument to be used as a reference instrument for the purposes of this Standard for measurement of transverse rail profile, the above results shall be documented and the primary profiles for each measurement shall be available for inspection by a Client.

A statement shall be made of the following:

- the temperature (or temperatures) at which measurements were undertaken for purposes of verification
- the temperature range over which the supplier of the test instrument believes that the verification is valid, if this is greater than the temperature range at which measurements were undertaken
- the reason(s) why the validation is acceptable for this temperature range
- the range of track gauges for which the instrument can be used as a reference instrument



Notes:

1. Dimensions are in mm and are approximate only
2. Profile to be measured after assembly
3. The beam shown is suitable for verifying instruments for wavelengths of less than 300 mm i.e. 10-30 mm, 30-100 mm and 100-300 mm.

Figure A.1 Schematic diagram of a calibration beam for longitudinal profile measuring equipment

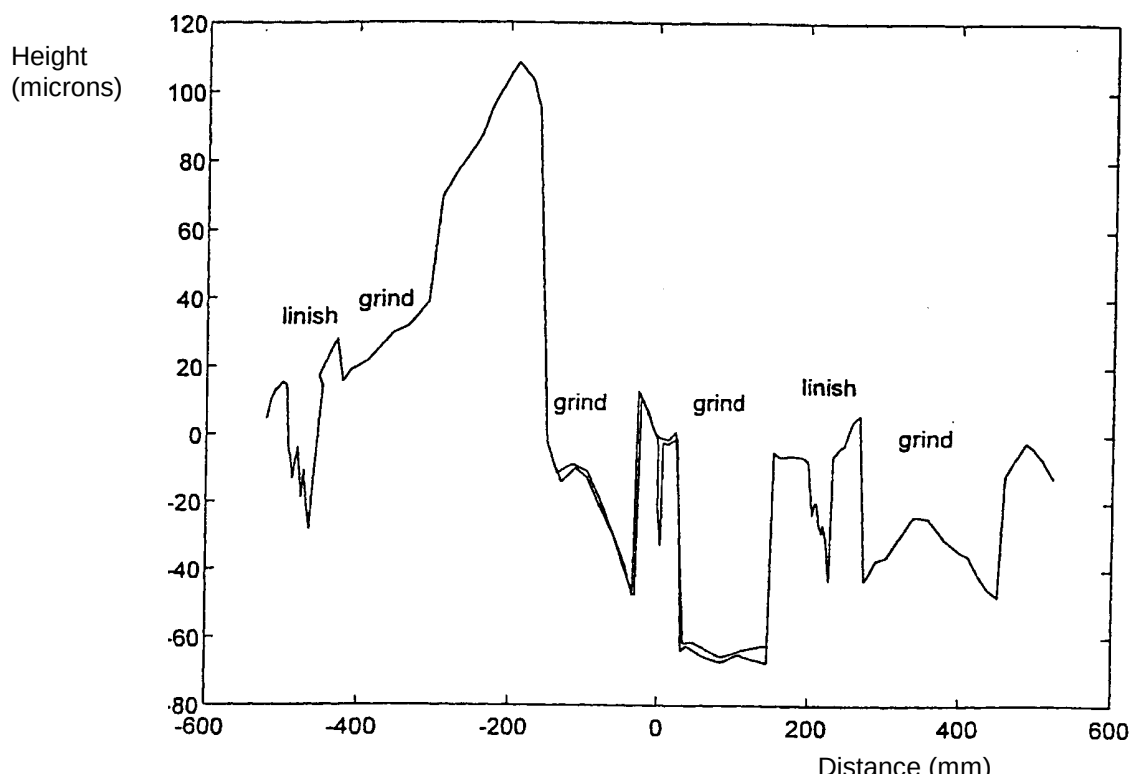


Figure A.2 Longitudinal profile of a calibration beam

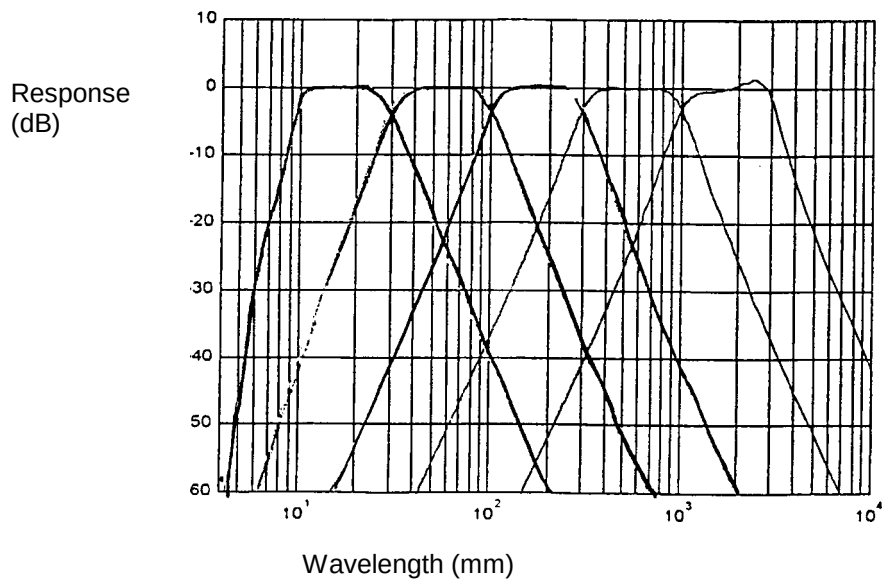
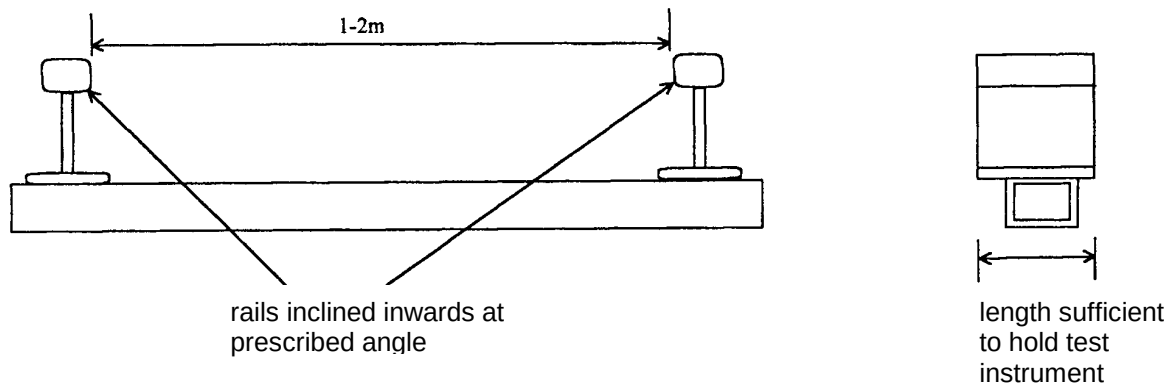


Figure A.3 Amplitude response of acceptable band-pass profile filters



Notes:

1. Profile of one rail section to differ by less than ± 0.3 mm from profile of reference rail
2. Profile of second rail section to differ by at least 0.5 mm at one or more points in the reprofiling zone.

Figure A.4 Schematic diagram of a calibration jig for verification of reference instruments for measuring the transverse rail profile.

Table A.4 (informative) Suggested format for tabulation of maximum positive and negative differences of CMM measurements of transverse profiles and of test instrument measurements of transverse profiles of rail sections on calibration jig (repeatability)

	A	B	C			A	B	C
A	0	+/-	+/-		A	0	+/-	+/-
B	+/-	0	+/-		B	+/-	0	+/-
C	+/-	+/-	0		C	+/-	+/-	0
first rail (difference less than +/-0.3mm from profile of reference rail)					second rail (difference greater than +/-0.5mm from profile of reference rail)			
	X	Y	Z			X	Y	Z
X	0	+/-	+/-		X	0	+/-	+/-
Y	+/-	0	+/-		Y	+/-	0	+/-
Z	+/-	+/-	0		Z	+/-	+/-	0

Table A.5 (informative) Suggested format for tabulation of maximum positive and negative differences of test instrument and CMM measurements of transverse profiles of rail sections on calibration jig

X	+/-	+/-	+/-		X	+/-	+/-	+/-
Y	+/-	+/-	+/-		Y	+/-	+/-	+/-
Z	+/-	+/-	+/-		Z	+/-	+/-	+/-
first rail (difference less than +/-0.3mm from profile of reference rail)					second rail (difference greater than +/-0.5mm from profile of reference rail)			

Annex B

(normative)

Procedures to demonstrate correlation of approved and reference instruments

B.1 Longitudinal profile

B.1.1 test sites

Measurements shall be taken at the following sites:

(T1) Tangent track which has been ground to the requirements of Class 1 in Section 4 of this Standard.

(L1) A left hand curve which has been ground to the requirements of Class 1 in Section 4 of this Standard. The nominal value of the curve radius shall not be greater than 800m. The nominal curve radius shall be recorded.

(R1) A right hand curve which has been ground to the requirements of Class 1 in Section 4 of this Standard. The nominal value of the curve radius shall not be greater than 800m. The nominal curve radius shall be recorded.

B.1.2 measurements required

For all sites, conformance with the Standard shall be demonstrated using a reference instrument.

The following measurements shall be taken at each of the test sites stated in Section B.1.1.

- longitudinal profile of both rails throughout the test site using a reference instrument
- longitudinal profile of both rails throughout the test site with the test instrument

Measurements shall be taken through the grinding site for a continuous length of at least 200m. If either the reference instrument or the test instrument requires a transition length at either the beginning or the end of the measurement, the transition length shall lie outside this 200m length.

NOTE - If the test instrument is required also to provide information on rail condition prior to grinding, it is recommended that a comparison such as that described here is made for measurements on test sites which do not satisfy the requirements of Class 1 in Section 4 of this Standard. Such test sites could be the same as sites T1, L1 and R1 before grinding is undertaken.

If the test instrument is to be used at a single speed, conformance shall be demonstrated for measurements taken with the instrument at that speed. If the test instrument is to be used for a range of speeds, conformance shall be demonstrated for the full range of speeds. In this case, measurements with the test instrument shall be made at the maximum speed on all of the test sites stated in Section B.1.1. Measurements shall also be made on test site T1 at the minimum speed. In all cases, the speed at which measurements are taken shall be recorded.

The ambient temperature and weather conditions during the tests shall be recorded. The direction of measurement with the reference instrument and the test instrument shall also be recorded. If conformance of the test instrument is to be demonstrated for both directions of measurement, all of the measurements described in this Section shall be taken in both directions on each of the three sites stated in Section B.1.1.

The conditions for which tests shall be undertaken in order to correlate results from the test instrument with those from reference instrument are shown in Table B.1 for the following sets of conditions in which an approved instrument is to be used:

- (a) at a single speed and in a single direction of travel
- (b) over a range of speeds and in a single direction of travel
- (c) at a single speed and in both directions of travel
- (d) over a range of speeds and in both directions of travel

If the acceptance criteria for longitudinal profile are stated in terms of a moving average of peak-to-peak amplitudes in the wavelength ranges stated in Section 4 of this Standard, the primary profiles for the reference instrument and for the test instrument shall be processed to give moving averages of peak-to-peak amplitudes for each of those wavelength ranges throughout the test site. If instead the acceptance criteria for longitudinal profile are stated in terms of a moving average of root-mean-square (RMS) amplitudes, these primary profiles shall be processed to give moving averages of RMS amplitudes. The window length for each wavelength range shall be as stated in Section 4 of this Standard.¹¹ The sampling interval and the digitisation increment shall be recorded.

¹¹ If a digital representation of the profile is taken, it is recommended that digitisation is undertaken with the same sampling interval and digitisation increment for both the reference instrument and the test instrument and that the signal processing procedure and filters are identical for all measurements.

NOTES

- 1 It is recommended that the sampling interval be no more than 2mm and the digitisation increment be no more than 1 micron.
- 2 In order to minimise testing, it is recommended that the same primary profiles be used to demonstrate satisfactory correlation of the test equipment for moving averages of both RMS and peak-to-peak amplitudes.
- 3 If a comparison is being made of moving averages of peak-to-peak amplitudes, it would in general be necessary to use phase correct profile filters to filter primary profiles from both the reference and the test instruments. In general, this is not necessary for a correlation based on moving averages of RMS amplitudes.

For each filtered profile, the percentage exceedence shall be calculated as a function of amplitude for the pertinent RMS or peak-to-peak moving average. These percentage exceedences shall be calculated for at least the values shown in Table B.2 (a) and (b) for RMS and peak-to-peak moving averages respectively.

There shall thus be a percentage exceedence as a function of amplitude for the reference and test instruments for a total of 12, 16, 24 and 32 combinations respectively of test site, measuring speed and wavelength range for the conditions shown in Notes a-d of Table B.1.

NOTE - The percentage exceedence as a function of amplitude is essentially a cumulative probability distribution for the appropriate RMS or peak-to-peak average of the filtered profile.

Table B.1 Set of test conditions for which correlation shall be demonstrated

test site	minimum measuring speed, v1	maximum or single measuring speed, v2
tangent, T1		
- leading	d	c,d
- trailing	d	c,d
- single direction	b	a,b
left hand curve, L1		
- leading	-	c,d
- trailing	-	c,d
- single direction	-	a,b
right hand curve, R1		
- leading	-	c,d
- trailing	-	c,d
- single direction	-	a,b

Note - a = set of conditions for an instrument which is to be used in a single direction at a single speed

b = set of conditions for an instrument which is to be used in a single direction for a range of speeds

c = set of conditions for an instrument which is to be used in both directions of travel for a single speed

d = set of conditions for an instrument which is to be used in both directions for a range of speeds

Table B.2(a) Values of moving average of RMS amplitude for which percentage exceedence shall be calculated for each test condition

wavelength range (mm)	amplitude (microns) for which percentage exceedence shall be calculated			
10-30	2	4	6	8
30-100	2	4	6	8
100-300	6	12	18	24
300-1000	20	40	60	80

Table B.2(b) Values of moving average of peak-to-peak amplitude for which percentage exceedence shall be calculated for each test condition

wavelength range (mm)	amplitude (microns) for which percentage exceedence shall be calculated			
10-30	5	10	15	20
30-100	5	10	15	20
100-300	15	30	45	60
300-1000	50	100	150	300

B.1.3 demonstration of satisfactory correlation; use of test instrument as an approved instrument

The following criteria shall be satisfied:

(1) The test instrument shall not consistently overestimate the percentage of track whose longitudinal profile complies with the requirements of rail ground to Class 1 in Section 4 of this Standard. This criterion shall be satisfied for each wavelength range if the test instrument overestimates the percentage of rail complying with the Standard for the following cases¹²

- for an instrument to be used in a single direction at a single speed: none of the test conditions shown by Note a of Table B.1
- for an instrument to be used in a single direction for a range of speeds: none of the test conditions shown by Note b of Table B.1
- for an instrument to be used in both directions of travel at a single speed: a maximum of one of the test conditions shown by Note c of Table B.1
- for an instrument to be used in both directions of travel for a range of speeds: a maximum of one of the test conditions shown by Note d of Table B.1 at speed v1 and one of the test conditions at speed v2

(2) The percentage exceedence for the two instruments shall be similar. This criterion shall be deemed to be satisfied if the percentage exceedence for the test instrument for each of the amplitudes in Table B.2(a) or B.2(b) is within +/-50% of the corresponding percentage exceedence for the reference instrument.

¹² The purpose of this requirement is to ensure that in the worst case, an approved instrument overestimates amplitudes of residual irregularities on the rail. Therefore if the approved instrument is used to demonstrate conformance with the criteria stated in Section 4 of this Standard, the amplitude of residual irregularities on the ground rail will be equal to or less than given in Section 4 of this Standard.

For each set of parameters in which these conditions are satisfied, the test instrument shall be approved. The test instrument shall not be approved if the requirements stated here are not demonstrated. The test instrument shall also not be approved for values of the operating parameters outside the range for which correlation has been demonstrated.

For an instrument which is to be used to demonstrate conformance to grinding Class 2 track only, it is necessary for the above criteria to be satisfied only for the 30-100mm and 100-300mm wavelength ranges.

B.1.4 Documentation

The results of this test of correlation shall be documented in tabular and/or graphical format. A statement shall be made of the following parameters for which satisfactory correlation of results from the test instrument is demonstrated:

- speed or range of speed
- direction of measurement
- wavelength ranges
- Class of grinding for which the instrument can be used

The ambient temperature and weather conditions during the tests shall be documented. If there are reasons to believe that the test instrument would behave significantly differently in different temperatures and weather conditions, these shall be stated. The test document and corroborative evidence, including primary and filtered profiles from both test and reference instruments, shall be available for inspection by a Client.

B.2 transverse profile

B.2.1 test sites

Measurements shall be taken at the following sites:

(T1) Tangent track which has been ground to the requirements of Class Q in Section 5 of this Standard. The length of the site shall be at least 500m.

(L1) A left hand curve which has been ground to the requirements of Class Q in Section 5 of this Standard. The nominal value of the curve radius, which shall not be greater than 800m, shall be recorded. The length of the site shall be at least 100m.

(R1) A right hand curve which has been ground to the requirements of Class Q in Section 5 of this Standard. The nominal value of the curve radius, which shall not be greater than 800m, shall be recorded. The length of the site shall be at least 100m.

B.2.2 measurements required

For these tests, the three ranges of deviation in Table 3 of this Standard shall be interpreted as +/- 0.3mm, +/-0.5mm and +/-0.85mm. Conformance with the Standard shall be demonstrated using a reference instrument.

The following measurements shall be taken at each test site:

- transverse profile of both rails throughout the test using a reference instrument
- transverse profile of both rails throughout the test site with the test instrument

Measurements shall be taken through the grinding site for a length of at least 500m. If either the reference instrument or the test instrument requires a transition length at either the beginning or the end of the measurement, the transition length shall lie outside this 500m length.

NOTE - If the test instrument is to be used also to assist in the grinding process, it is recommended that a comparison such as that described here is used for measurements on test sites which do not satisfy the requirements of Class Q in Section 5 of this Standard. Such test sites could be the same as sites T1, L1 and R1 before grinding is undertaken.

For both the reference and test instruments, at least 50 measurements shall be recorded on each rail. The distance along the track between successive recordings shall be no more than 20m and no less than 2m. The ambient temperature and weather conditions during the tests shall be recorded. The test instrument shall be set up so that the system operates in the mode in which it is normally used to provide transverse profile information for the ground track e.g. if individual measurements of the transverse profile are usually provided, these shall be used for the correlation described here; alternatively if averaged profiles are usually provided, these averaged profiles shall be used for the correlation. This test set up shall be recorded.

The conditions for which measurements shall be made with the test instrument are shown in Table B.1 for four sets of conditions in which correlation of the test instrument with the reference instrument is to be demonstrated.

The maximum positive and negative deviations of each measured profile from the appropriate reference rail shall be found as stated in Section 5.1 of this Standard. For the reference instrument, the percentage of measurements shall be calculated for each test site and for each rail which lie within the following ranges of deviation: $\pm 0.1\text{mm}$, $\pm 0.2\text{mm}$, $\pm 0.3\text{mm}$, $\pm 0.4\text{mm}$, $\pm 0.5\text{mm}$, $\pm 0.6\text{mm}$, $\pm 0.7\text{mm}$, $\pm 0.85\text{mm}$, $\pm 1.0\text{mm}$ ¹³ For the test instrument, these percentages shall be calculated for each test site, rail, measuring direction and speed of travel.

B.2.3 demonstration of satisfactory correlation; use of test instrument as an approved instrument

The following criteria shall be satisfied:

- (1) The test instrument shall not overestimate the percentage of track whose transverse profile complies with the requirements of rail ground to Class Q in Section 5 of this Standard. This criterion shall be satisfied if the test instrument overestimates the percentage of rail complying with the Standard for the following cases¹⁴
- (2) for an instrument to be used in a single direction at a single speed: none of the test conditions shown by Note a in Table B.1
 - for an instrument to be used in a single direction for a range of speeds: none of the test conditions shown by Note b in Table B.1
 - for an instrument to be used in both directions of travel at a single speed: a maximum of one of the test conditions shown by Note c in Table B.1
 - for an instrument to be used in both directions of travel for a range of speeds: a maximum of one of the test conditions shown by Note d in Table B.1 at speed v_1 and one of the test conditions at speed v_2
- (3) The percentage exceedences for the two instruments shall be similar. This criterion shall be deemed to be satisfied if the percentage exceedence for the test instrument for each

¹³ These include the three ranges stated in Section 5.2 of this Standard.

¹⁴ The purpose of this requirement is to ensure that in the worst case, an approved instrument overestimates amplitudes of residual irregularities on the rail. Therefore if the approved instrument is used to demonstrate conformance with the criteria stated in Section 4 of this Standard, the amplitude of residual irregularities on the ground rail will be equal to or less than given in Section 4 of this Standard.

range of deviation is within +/-50% of the corresponding percentage exceedence for the reference instrument.

For each set of parameters in which these conditions are satisfied, the test instrument shall be approved. The test instrument shall not be approved if the requirements stated here are not demonstrated. The test instrument shall also not be approved for values of the operating parameters outside the range for which correlation has been demonstrated.

B.2.4 documentation

The results of this test of correlation shall be documented in tabular and/or graphical format. A statement shall be made of the following parameters for which satisfactory correlation of results from the test instrument is demonstrated:

- speed or range of speed
- direction of measurement
- Class of grinding for which the instrument can be used

The ambient temperature and weather conditions during the tests shall be documented. If there are reasons to believe that the test instrument would behave significantly differently in different temperatures and weather conditions, these shall be stated.

The test document and corroborative evidence, including records of the transverse profiles from both test and reference instruments, shall be available for inspection by a Client.

B.3 surface roughness

Instruments to measure surface roughness shall conform with the requirements of ISO 3274:1998¹⁵

B.4 metal removal

Reference instruments shall be used to measure metal removal.

¹⁵ ISO 3274 sets out nominal characteristics of instruments for measuring surface texture. ISO 3274:1988 indicates (in Note 1 of Scope on p1 of the Standard) that calibration procedures for such instruments will be introduced in a future Standard. This is not available at the time of writing the present document ().

Annex C
(informative)

Example of acceptance documentation for grinding work

- see German acceptance form -

Bahn _____

Abnahmeniederschrift für Oberbauleistungen

Oberbauleistungen _____

Strecke/Bahnhof _____

Gleis von _____ nach _____

von km _____ nach km _____

Weichen Nr. _____

Auftraggeber (zuständige Stelle) _____

Auftragnehmer (Firma) _____

Auftrag Nr. _____ vom _____

Die Leistungen wurden ausgeführt im Zeitraum von _____ bis _____

Der Abnahme liegen folgende Dokumente zugrunde:

Anwesend: Für den Auftraggeber (Bahn):

Für den Auftragnehmer (Firma):

Ergebnis

Die Leistungen wurden abgenommen. *)

Die Leistungen wurden abgenommen; jedoch wurden die auf der Rückseite aufgeführten Mängel festgestellt. *)

Die Mängel sind bis zum _____ zu beseitigen.
Datum

Ort, Datum

Für den Auftragnehmer (Firma):

Für den Auftraggeber (Bahn):

Die Beseitigung der umseitig aufgeführten Mängel wird bescheinigt:

Unterschrift und Datum des Beauftragten des Auftraggebers

*) Nichtzutreffendes streichen